Causal Intervention and Counterfactual Reasoning for Multi-modal Fake News Detection

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Abstract

Due to the rapid upgrade of social platforms, most of today's fake news is published and spread in a multi-modal form. Most existing multi-modal fake news detection methods neglect the fact that some label-specific features learned from the training set cannot generalize well to the testing set, thus inevitably suffering from the harm caused by the latent data bias. In this paper, we analyze and identify the psycholinguistic bias in the text and the bias of inferring news label based on only image features. We mitigate these biases from a causality perspective and propose a **C**ausal intervention and **C**ounterfactual reasoning based **D**ebiasing framework (CCD) for multi-modal fake news detection. To achieve our goal, we first utilize causal intervention to remove the psycholinguistic bias which introduces the spurious correlations between text features and news label. And then, we apply counterfactual reasoning by imagining a counterfactual world where each news has only image features for estimating the direct effect of the image. Therefore we can eliminate the image-only bias by deducting the direct effect of the image from the total effect on labels. Extensive experiments on two real-world benchmark datasets demonstrate the effectiveness of our framework for improving multi-modal fake news detection.

1 Introduction

Fake news quietly sneaks into people's daily life, mixed with massive information, causing serious impact and harm to society. Fake news often utilizes multimedia information such as text and images to mislead readers, spreading and expanding its influence. Thus, it is crucial and urgent to find a way to discern multi-modal fake news.

Today, most existing methods train on known fake news instances expecting to capture the label-specific features for judging the authenticity of unseen news (Singhal et al., 2020; Wu et al., 2021; Qian et al., 2021b; Qi et al., 2021). However, such label-specific features may expose the models to hidden data bias when confronted with unseen fake news samples (Wang et al., 2018; Cheng et al., 2021; Zhu et al., 2022). To address the problem, we investigate the biases underlying the multi-modal fake news detection data and identify the psycholinguistic bias in the text and the bias of inferring news label based on image features only (i.e. image-only bias). These biases could lead to spurious correlations between the news and labels, thus impairing the model performance on testing data.

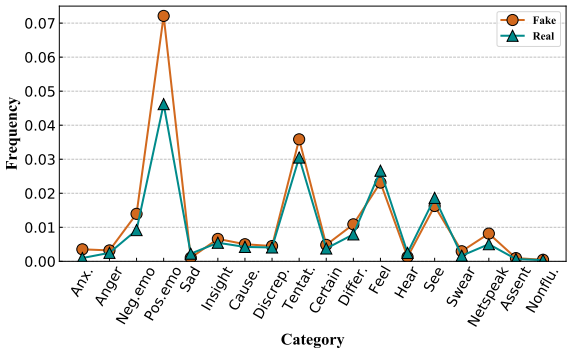


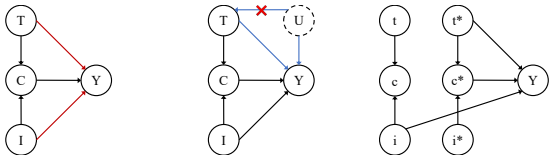
Figure 1: The word frequency distributions of psychological categories on different labels (Twitter dataset).

To explicitly explain the biases, we first formulate the process of fake news detection as a causal graph as shown in Figure 2(a). In addition to the impact of fused features C on news label Y that most multi-modal fake news detection methods focus on, other two edges are pointing to Y, starting from text features T, and image features I, respectively. Generally speaking, the publishers of fake news would try their best to fabricate confusing text or use certain techniques to forge fake images. This makes the text and image can individually affect the news label.

For the $T \rightarrow Y$ branch, we observe that the linguistic characteristics of the text have obvious emotional preferences, such as the usage of psycholin-

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(a) The proposed causal graph (Factual world).

(b) Causal graph with confounder.

(c) Counterfactual world.

Figure 2: The causal graphs for fake news detection. T: text features, I: image features, C: multi-modal features (i.e., the fused features of image and text), Y: news label, U: confounder. *denotes the reference values.

guistic words "crazy" and "amazing", which play a critical role in fake news detection. To deeply analyze the linguistic characteristics of the text, we present a mathematical analysis of the psycholinguistic word distribution of real news and fake news based on the LIWC 2015 dictionary (Pennebaker et al., 2015). Take the Twitter dataset as an example, as shown in Figure 1, we can observe that the word frequency distribution of fake news is quite different from that of real news, especially for words expressing anxiety, negative emotions, positive emotions, tentative, and netspeak. It seems that we can draw a conclusion that fake news prefers to use loaded language to stir up the reader's emotions and attract more attention. Consequently, the model could be prone to relying on such psycholinguistic features as a shortcut to judge news authenticity. However, we analyze the training set and testing set, and find that there exist significant differences in the frequency of these psycholinguistic words. The manifest differences between the training set and testing set have proven that this shortcut appears to be unreliable evidence. As shown in Figure 2(b) where U denotes the confounder (i.e. the psycholinguistic features in the text), there exist a backdoor path $T \leftarrow U \rightarrow Y$ which will introduce spurious correlations among the text features and news label. In order to remove the psycholinguistic bias, we apply causal intervention by adopting the backdoor adjustment (Glymour et al., 2016) with do-calculus $P(Y|do(T))$ to calculate the causal effect in the training stage, which is fundamentally different from the conventional likelihood $P(Y|T)$.

For the $I \rightarrow Y$ branch, we observe from the datasets that two different news pieces sharing the same image could have contrary labels. This shows that sometimes even if the image is real, the text could be fabricated, and the news could thus be fake. We can take advantage of images as an additional modality to provide more detection evidence, but it is unreliable to infer the authenticity of the news based on the image features alone. In this case, we argue that the image-only bias (i.e., the direct causal effect from image features alone to news label) should be eliminated. Towards this end, we use counterfactual reasoning by imagining a counterfactual world (Figure 2(c)) where both text features T and fused features C are not given (represented by reference values t^* and c^*), except for image features I. In this way, the bias can be estimated by computing the direct causal effect of I on Y and we can conduct the debiasing by subtracting it from the total effect on Y.

We instantiate our proposed debiasing framework on three strong baseline models that can handle both text and image features as inputs. Extensive experiments on two widely used real-world benchmark datasets show the effectiveness of our framework. Overall, our contributions can be summarized as follows:

- We analyze each modality of fake news detection data and identify the underlying psycholinguistic bias in the text and the image-only bias. And we propose a novel **C**ausal intervention and **C**ounterfactual reasoning based **D**ebiasing framework (CCD) for multi-modal fake news detection.
- In our debiasing framework CCD, we conduct causal interventions via backdoor adjustment to remove spurious correlations introduced by the psycholinguistic confounder. For addressing the image-only bias, we apply counterfactual reasoning to pursue the indirect causal effect as the inference prediction.
- Our causal framework CCD can be applied to any fake news detection model with image and text features as inputs. We implement the proposed framework on three strong baseline models, and conduct extensive experiments on two widely used benchmark datasets, validating the effectiveness of CCD.

2 Preliminaries

2.1 Causal Graph

The causal graph (Glymour et al., 2016) is a probabilistic graphical model used to describe how variables interact with each other, expressed by a directed acyclic graph $\mathcal{G} = \{\mathcal{N}, \mathcal{E}\}$ consisting of the

sets of variables $\mathcal{N}$ and the causal correlations $\mathcal{E}$ between two nodes. As shown in Figure 3, $X \rightarrow Y$ denotes that X is the cause of the effect Y. U is the confounder.

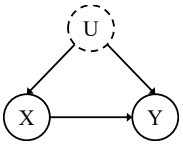


Figure 3: An example of causal graph.

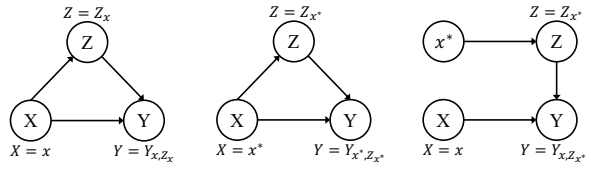
2.2 Causal Intervention

Causal intervention is used to seek the real causal effect of one variable on another when there exist confounders. In a causal graph, the intervening operation on a variable removes all edges pointing to it, such that its parent nodes no longer cause it. The backdoor adjustment (Glymour et al., 2016) with do-calculus offers a tool for calculating the intervened distribution under the condition of no extra confounders. For the example in Figure 3, the adjustment formula can be derived according to Bayes' theorem as follows, where u denotes the value of confounder U:

$$P(Y|do(X)) = \sum_u P(Y|X,u)P(u). \quad (1)$$

2.3 Counterfactual Reasoning and Causal Effect

Counterfactual reasoning (Pearl, 2009) is a statistical inference method used to infer outcomes under hypothetical conditions that are different from the factual world. By conducting counterfactual reasoning, we can estimate the causal effect (Pearl, 2022) of a treatment variable on a response variable. For instance, Figure 4 shows an abstract setting for estimating and removing the direct influence of X on Y. Figure 4(a) is the factual world where the calculation of Y is denoted as $Y_{x,Z_x} = Y(X = x, Z = Z(X = x))$.



(a) (b) (c)

Figure 4: Example of causal graph where X, Y, and Z denote cause, effect and mediator variable, respectively. *denotes the reference values.

Based on Figure 4(a) and 4(b), we define the total effect (TE) of $X = x$ on Y as:

$$\text{TE} = Y_{x,Z_x} - Y_{x^*,Z_{x^*}}, \quad (2)$$

which can be seen as the comparisons between two potential outcomes of X given two different treatments, i.e., $X = x$ and $X = x^*$. The total effect (TE) can be decomposed into the sum of the natural direct effect (NDE) and the total indirect effect (TIE), namely, TE = NDE + TIE. NDE represents the natural direct effect of X on Y when the mediator variable Z is blocked (Figure 4(c)):

$$\text{NDE} = Y_{x,Z_{x^*}} - Y_{x^*,Z_{x^*}}. \quad (3)$$

$Y_{x,Z_{x^*}}$ is calculated under the counterfactual world where X can be simultaneously set to different values x and x^* (Figure 4(c)). Thus, TIE (the total indirect effect of X on Y) can be obtained:

$$\text{TIE} = \text{TE} - \text{NDE} = Y_{x,Z_x} - Y_{x,Z_{x^*}}. \quad (4)$$

We use TIE as the debiased result for inference.

3 Method

In this section, we first formulate the fake news detection task as a causal graph to clearly depict the causal effect between factors. And then we present our CCD framework that removes the psycholinguistic bias by means of causal intervention, as well as deducts the direct causal effect of image features (i.e. the image-only bias) via counterfactual reasoning.

3.1 Causal Graph of Fake News Detection

As aforementioned, Figure 2(a) depicts the causal graph of the fake news detection process. Nodes T, I, and C represent the text features, image features, and fused multi-modal features, respectively. According to the proposed causal graph, the final prediction Y takes inputs from the three branches: the direct effect of the input T and I on Y via $T \rightarrow Y$ and $I \rightarrow Y$, as well as the indirect effect of the input T and I on Y via the fused features C, i.e. $T(I) \rightarrow C \rightarrow Y$. Each branch of Figure 2(a) can be implemented via a base fake news detection model (Figure 5). Formally, the abstract format of the model should be:

$$Y_{t,i,c} = Y(T = t, I = i, C = c), \quad (5)$$

where $c = f(T = t, I = i)$, $f(\cdot)$ is the feature aggregation function in baseline fake news detection

models. Then the total effect (TE) of the input on label y can be written as:

$$\text{TE} = Y_{t,i,c} - Y_{t^*,i^*,c^*}, \tag{6}$$

where t^* and i^* are respectively the reference values of T and I, and $c^* = f(T = t^*, I = i^*)$. As introduced in Section 2.3, the reference status is defined as the status of blocking the signal from text and image, i.e., t and i are not given (void values). For implementation, we use tensors filled with the scalar value 0 to represent the reference values t^* and i^*. In this way, the inputs do not contain any semantic information.

Following previous studies (Niu et al., 2021; Wang et al., 2021; Tian et al., 2022), we calculate the prediction $Y_{t,i,c}$ through a model ensemble with a fusion function.

$$\begin{aligned} Y_{t,i,c} &= Y(T = t, I = i, C = c) \\ &= \mathcal{F}(Y_t, Y_i, Y_c) \\ &= Y_c + tanh(Y_t) + tanh(Y_i), \end{aligned} \tag{7}$$

where Y_t is the output of the text-only branch (i.e. $T \rightarrow Y$), Y_i is the output of the image-only branch (i.e. $I \rightarrow Y$), and $Y_c = Y_{t,i}$ is the output of fused features branch (i.e. $C \rightarrow Y$) as shown in Figure 5. $\mathcal{F}(\cdot)$ is the fusion function to obtain the final prediction. We adopt a non-linear fusion strategy for its better representation capacity (Wang et al., 2021). Any differentiable arithmetic binary operations can be employed as the fusion function $\mathcal{F}(\cdot)$ and we examine several fusion alternatives in Table 4.

3.2 Deconfounded Training with Causal Intervention

As Figure 2(b) shows, there exist an unobserved confounder U (i.e., the psycholinguistic of the text) in the $T \rightarrow Y$ branch, which causes spurious correlations between the text features and news label by learning the likelihood $P(Y|T)$. In order to explicitly illustrate the impact of the confounder, we use Bayes' theorem:

$$\begin{aligned} P(Y|T) &= \textstyle\sum_u P(Y|T,u)P(u|T) \\ &\propto \textstyle\sum_u P(Y|T,u)P(T|u)P(u). \end{aligned} \tag{8}$$

Next, we conduct deconfounded training in $T \rightarrow Y$ branch which exploits the backdoor adjustments (Glymour et al., 2016) with do-calculus on T to calculate the corresponding intervention distribution. Since the edge $U \rightarrow T$ has been cut off, we can have:

$$\begin{aligned} Y_t &= P(Y|do(T)) \\ &= \textstyle\sum_u P(Y|T,u)P(u). \end{aligned} \tag{9}$$

To estimate Y_t, given the text features T's representations $\mathbf{t}$ and the confounder U's representations $\mathbf{u}$, Equation (9) is implemented as $\sum_{\mathbf{u}} P(y|\mathbf{t},\mathbf{u})P(\mathbf{u})$, where $P(y|\mathbf{t},\mathbf{u})$ is the prediction upon a news feature learning model $g(\cdot)$:

$$P(y|\mathbf{t},\mathbf{u}) = \sigma(g(\mathbf{t},\mathbf{u})), \tag{10}$$

where $\sigma(\cdot)$ is the sigmoid function that forms the output of $g(\cdot)$ into $(0, 1)$. In summary, the implementation of Equation (9) is formally defined as:

$$\begin{aligned} P(Y|do(T)) &= \mathbb{E}_u[P(Y|T,u)] \\ &= \mathbb{E}_u[\sigma(g(\mathbf{t},\mathbf{u}))]. \end{aligned} \tag{11}$$

Note that $\mathbb{E}_u$ requires expensive sampling. Following recent works (Wang et al., 2020; Yang et al., 2021), we can apply Normalized Weighted Geometric Mean (NWGM) (Xu et al., 2015) to approximate the above expectation by moving the outer expectation into the sigmoid function as:

$$P(Y|do(T)) \overset{\text{NWGM}}{\approx} \sigma(\mathbb{E}_u[g(\mathbf{t},\mathbf{u})]). \tag{12}$$

We apply a linear model to approximate the conditional probability, i.e. the probability of Y under the conditions T and U. Inspired by previous works (Chen et al., 2022a; Tian et al., 2022), we model $g(\mathbf{t},\mathbf{u}) = \mathbf{W}_t\mathbf{t} + \mathbf{W}_u \cdot h(\mathbf{u})$, where $h(\mathbf{u})$ is the feature transformation of $\mathbf{u}$, $\mathbf{W}_t$ and $\mathbf{W}_u$ are learnable weight parameters. In this case, $\mathbb{E}_u[g(\mathbf{t},\mathbf{u})] = \mathbf{W}_t\mathbf{t} + \mathbf{W}_u \cdot \mathbb{E}_u[h(\mathbf{u})]$.

To compute $\mathbb{E}_u[h(\mathbf{u})]$, we implement $h(\mathbf{u})$ as the scaled Dot-Product attention (Vaswani et al., 2017). We resort to LIWC 2015 dictionary (Pennebaker et al., 2015) to approximate U as a fixed confounder dictionary $\mathbf{D}_u = [\mathbf{u}_1, \mathbf{u}_2, ..., \mathbf{u}_N] \in \mathbb{R}^{N \times d_u}$, where N is the number of word categories and d_u is the hidden feature dimension. Then we have:

$$\mathbb{E}_u[h(\mathbf{u})] = \textstyle\sum_u [softmax(\frac{\mathbf{Q}^T\mathbf{K}}{\sqrt{d_m}}) \odot \mathbf{D}_u]P(\mathbf{u}), \tag{13}$$

where $\mathbf{Q} = \mathbf{W}_q\mathbf{t}$, $\mathbf{K} = \mathbf{W}_k\mathbf{D}_u$ ($\mathbf{W}_q$ and $\mathbf{W}_k$ are learnable weight parameters), d_m denotes the scaling factor. $P(\mathbf{u})$ denotes the prior statistic probability and $\odot$ is the element-wise product.

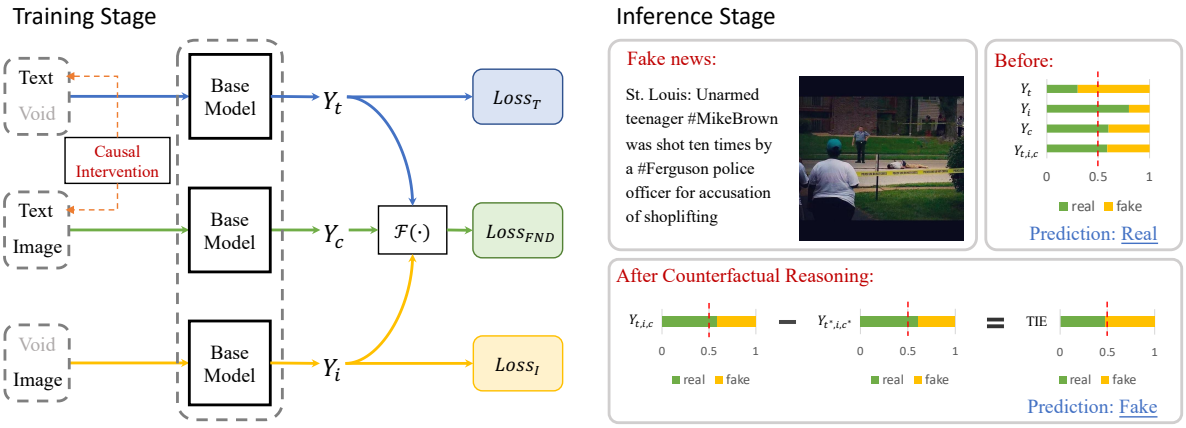


Figure 5: Illustration of the training and inference of our proposed CCD framework.

3.3 Mitigating the Image-only Bias with Counterfactual Reasoning

So far, the psycholinguistic bias has been successfully removed in the $T \rightarrow Y$ branch, but the fake news detection model based on the causal graph in Figure 2(a) still suffers from the image-only bias. This is because the prediction, i.e., $Y_{t,i,c}$, is still affected by the direct effect of the image. Consequently, fake news with more convincing image features still achieves a high probability of being judged as real news. To mitigate the image-only bias, we propose counterfactual reasoning to estimate the direct causal effect of I on Y by blocking the impact of T and C. Figure 2(c) shows the causal graph of the counterfactual world for fake news detection which describes the scenario when I is set to different values i and i^*. We also set T to its reference value t^*, therefore C would attain the value c^* when $T = t^*$ and $I = i^*$. In this way, the inputs of T and C are blocked, and the model can only rely on the given image i for detection. We can thus obtain the natural direct effect (NDE) of I on Y, namely the image-only bias:

$$\text{NDE} = Y_{t^*,i,c^*} - Y_{t^*,i^*,c^*}. \quad (14)$$

Furthermore, the removal of the bias can be realized by subtracting NDE from the total effect TE:

$$\text{TIE} = \text{TE} - \text{NDE} = Y_{t,i,c} - Y_{t^*,i,c^*}. \quad (15)$$

TIE is the debiased result we used for inference.

3.4 Training and Inference

We illustrate the training and inference of our proposed CCD framework in Figure 5. Following Wang et al. (2021); Niu et al. (2021); Tian et al. (2022), for the training stage, we compute the loss for each branch, including the base multi-modal fake news detection branch ($Loss_{FND}$), the text-only detection branch ($Loss_T$), and the image-only detection branch ($Loss_I$). As such, we minimize a multi-task training objective to learn the model parameters, which is formulated as:

$$Loss = Loss_{FND} + \alpha Loss_T + \beta Loss_I, \quad (16)$$

where the loss $Loss_{FND}$ refers to the cross-entropy loss associated with the predictions of $\mathcal{F}(Y_t, Y_i, Y_c)$ from Equation (7). The text-only and image-only loss $Loss_T$ and $Loss_I$ are cross-entropy losses associated with the predictions of Y_t and Y_i. α and β are the trade-off hyperparameters.

In the inference stage, we use the de-biased effect for inference, which is implemented as:

$$\text{TIE} = Y_{t,i,c} - Y_{t^*,i,c^*} \quad (17)$$

$$= \mathcal{F}(Y_t, Y_i, Y_c) - \mathcal{F}(Y_{t^*}, Y_i, Y_{c^*}). \quad (18)$$

4 Experiments

In this section, we apply our CCD framework on three strong baseline multi-modal fake news detection models on two real-world datasets to evaluate the effectiveness of our proposed CCD framework.

4.1 Experimental Settings

4.1.1 Datasets

We conducted experiments on two datasets:

Twitter: This dataset was released for Verifying Multimedia Use task at MediaEval[1]. It consists of tweets with textual, visual, and social context information. Since our framework belongs to content-based methods, we only leverage textual and visual information.

[1]http://www.multimediaeval.org/mediaeval2015/.

News	Twitter	Pheme
# of Real News	7898	1972
# of Fake News	6026	3830
# of Images	514	3670

Table 1: The statistics of two real-world datasets.

Pheme: This dataset was generated as part of the Pheme project, which attempts to detect and verify rumors spread via social media. It is based on five breaking news stories, each of which comprises a series of statements categorized as rumor or non-rumor. We classified rumors as fake news and non-rumors as real news in our framework.

Our data preprocessing and division of the training set and testing set for both datasets are the same as previous work (Qian et al., 2021b). Table 1 shows the statistics of the two datasets.

4.1.2 Base Models

The CCD framework can be applied to any multi-modal fake news detection method with text and image as input. Here, we apply our framework to the following strong baselines: 1) **SpotFake+** (Singhal et al., 2020): SpotFake+ concatenates the features extracted from different modalities and performs multiple feature transformations to facilitate multi-modal fusion. 2) **MCAN** (Wu et al., 2021): MCAN stacks multiple co-attention layers to learn dependencies across the modalities. They repeatedly fuse the two modalities to simulate people's reading process. 3) **HMCAN** (Qian et al., 2021b): HMCAN uses a hierarchical multi-modal contextual attention model that considers both the text's hierarchical semantics and multi-modal contextual data.

4.1.3 Evaluation Metrics

We use the *Accuracy* as the evaluation metric for binary classification tasks such as fake news detection. In consideration of the imbalance label distributions, in addition to the accuracy metric, we add *Precision*, *Recall*, and *F1-score* as complementary evaluation metrics following previous works (Wu et al., 2021; Qian et al., 2021b).

4.1.4 Implementation Details

All of the methods are trained for 200 epochs and the initial learning rate for the Adam optimizer is tuned in [1e-5, 1e-3]. For the confounder dictionary $\mathbf{D}_u \in \mathbb{R}^{N \times d_u}$, N is 18 (Anger, Anxiety, Assent, Causation, Certainty, Differentiation, Discrepancy, Feel, Hear, Insight, Negative emotion, Netspeak, Nonfluencies, Positive emotion, Sadness, See, Swear words, Tentative), and d_u is set to 4. For the scaled Dot-Product attention, the scaling factor d_m is set to 256. As for other necessary hyperparameters in the baseline methods, our settings are consistent with them.

4.2 Experimental Results

Table 2 displays the experimental results of our proposed framework CCD applied to the baseline methods on two benchmark datasets. The results of the baselines are the results of our reproductions on our data settings based on their public code[2]. From Table 2, we can obtain the following observations:

Compared with each base fake news detection model (i.e. SpotFake+, MCAN, HMCAN), the accuracy of the models that apply the proposed CCD framework (i.e., w/ CCD) has been significantly improved by around 7.7%, 3.3%, and 5.2% on the Twitter dataset, and improved by around 1.0%, 0.6%, and 1.3% on the Pheme dataset. With the help of the proposed framework, all of the base models show significant improvements on most metrics, which demonstrates the effectiveness of the proposed framework. We believe that CCD benefits from the removal of psycholinguistic bias with causal intervention as well as the mitigation of the image-only bias via counterfactual reasoning.

The performance improvements on the Twitter dataset are larger than that on the Pheme dataset. We attribute such a difference between the two datasets to the following two reasons: 1) The proportion of psycholinguistic vocabulary in the Twitter dataset (19.87%) is higher than that in the Pheme dataset (16.19%), so the Twitter dataset could be more susceptible to psycholinguistic bias. 2) According to Table 1, the number of unique images in the Twitter dataset is far less than the number of news texts, which means that there's a serious problem of different texts sharing the same image. So the influence of image-only bias in the Twitter dataset is more severe than that of the Pheme dataset.

4.3 Ablation Study of Causal Inference

We conduct experiments to study the de-biasing effect of each module in CCD using the strong baseline HMCAN on Twitter and Pheme testing

[2]https://github.com/shiivangii/SpotFakePlus.
https://github.com/wangjinguang502/HMCAN.
https://github.com/wuyang45/MCAN_code.

Dataset	Methods	Accuracy	Fake news			Real news		
			Precision	Recall	F1	Precision	Recall	F1
Twitter	SpotFake+	0.795	0.622	0.607	0.614	0.856	0.864	0.860
	w/ CCD	**0.856***	0.750	0.849	**0.797***	0.920	0.860	**0.889***
	MCAN	0.799	0.980	0.401	0.569	0.770	0.996	0.869
	w/ CCD	**0.825***	0.829	0.595	**0.692***	0.824	0.939	**0.878***
	HMCAN	0.831	0.955	0.514	0.668	0.804	0.988	0.887
	w/ CCD	**0.874***	0.820	0.792	**0.806***	0.899	0.914	**0.906***
Pheme	SpotFake+	0.815	0.711	0.525	0.604	0.840	0.921	0.879
	w/ CCD	**0.823***	0.714	0.574	**0.636***	0.854	0.915	**0.883***
	MCAN	0.834	0.716	0.639	0.675	0.872	0.906	0.889
	w/ CCD	**0.839***	0.693	0.721	**0.707***	0.896	0.882	**0.889**
	HMCAN	0.848	0.762	0.705	**0.732**	0.881	0.908	0.894
	w/ CCD	**0.859***	0.764	0.689	0.724	0.889	0.921	**0.905***

Table 2: Results of comparison among different models on Twitter and Pheme datasets. The best results are in bold. The marker * indicates that the improvement is statistically significant compared with the baseline (t-test with p-value < 0.05).

Dataset	Method	Accuracy
Twitter	HMCAN w/CCD	**0.874**
	w/o CI	0.842
	w/o CR	0.855
Pheme	HMCAN w/ CCD	**0.859**
	w/o CI	0.852
	w/o CR	0.850

Table 3: Impact of Causal Inference

Strategy	Accuracy	$F1_{Fake}$	$F1_{Real}$
MUL-sigmoid	0.695	0.569	0.765
MUL-tanh	0.733	0.472	0.821
SUM-sigmoid	0.806	0.600	0.872
SUM-tanh	**0.859**	**0.724**	**0.905**

Table 4: Impact of Different Fusion Strategies.

set. As shown in Table 3, we test the performance of CCD removing the causal intervention part (w/o CI), and CCD removing the counterfactual reasoning part (w/o CR). The variant model (w/o CI) does not consider the psycholinguistic confounder and uses the original text features for detection. While the variant model (w/o CR) uses $Y_{t,i,c}$ for inference without subtracting the direct effect of the image. We can observe that if we remove the causal intervention part, the performance respectively drops by around 3.7% and 0.8% on Twitter and Pheme, demonstrating the effectiveness of eliminating the psycholinguistic bias in the text. And removing the counterfactual reasoning part will make the performance respectively decreases by around 2.2% and 1.0% on Twitter and Pheme, proving that CCD can effectively mitigate the image-only bias in the inference stage.

4.4 Impact of Different Fusion Strategies

Following prior studies (Wang et al., 2021), we devise several differentiable arithmetic binary operations for the fusion strategy in Equation (7):

$$\begin{cases} \text{MUL-sigmoid} : Y_{t,i,c} = Y_c * \sigma(Y_t) * \sigma(Y_i), \\ \text{MUL-tanh} : Y_{t,i,c} = Y_c * tanh(Y_t) * tanh(Y_i), \\ \text{SUM-sigmoid} : Y_{t,i,c} = Y_c + \sigma(Y_t) + \sigma(Y_i), \\ \text{SUM-tanh} : Y_{t,i,c} = Y_c + tanh(Y_t) + tanh(Y_i). \end{cases} \tag{19}$$

The performance of different fusion strategies are reported in Table 4. From the table, we can find that SUM-tanh achieves the best performance over the other fusion strategies. This shows that a fusion function with the proper boundary is suitable for CCD. Multiple fusion strategies are worth studying when CCD is applied to other scenarios in the future.

4.5 Impact of the Value of α and β

We tune the trade-off hyperparameters α and β in the training objective by grid search in $\{0, 0.1, 0.25, 0.5, 0.75, 1, 2, 3, 4, 5\}$. And we find out that when α = 3 and β = 0.1, we can obtain satisfactory results in terms of accuracy on both datasets. To evaluate the impact of each parameter on the detection performance, we further study the accuracy under different values of α and β individually by fixing the other hyperparameter on

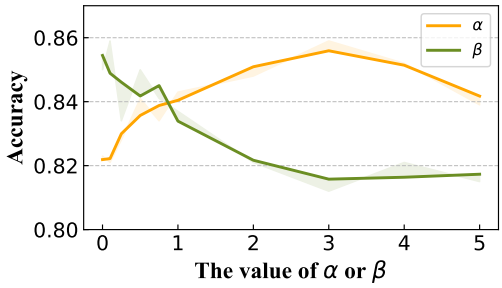


Figure 6: Impact of the Value of α and β

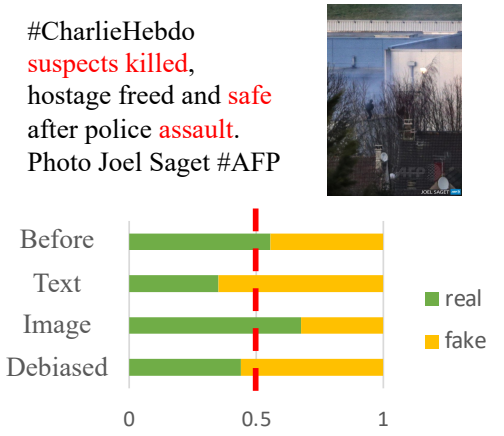
the Pheme dataset. As shown in Figure 6, when β=0.1 and α grows from 0 to 3, the accuracy keeps raising, indicating the importance of leveraging the text features that have removed psycholinguistic bias. When α=3 and β grow from 0 to 0.1, the accuracy increases, indicating the importance of capturing image-only bias. However, when α>3 or β>0.1, the performance decreases. It is because the training loss of the detection model using multi-modal features will be less important, which brings worse results.

4.6 Case Study

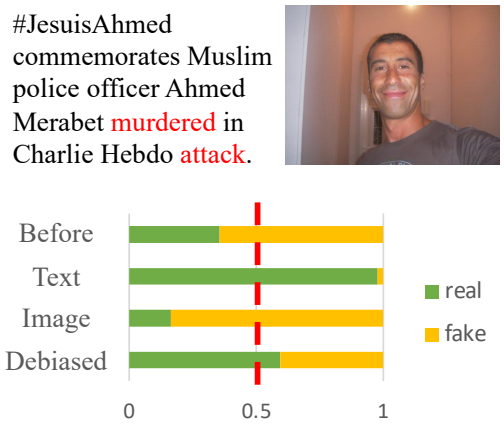
We provide a qualitative analysis of the proposed CCD framework by examining the fake and real news samples that are successfully detected by HMCAN w/ CCD on Pheme datasets in Figure 7. The psycholinguistic words are highlighted in red and the prediction results before (Before) and after (Debiased) counterfactual reasoning are shown in the charts. As we can see, the texts of both fake and real news contain words expressing anger and negative emotions (i.e., "killed", "assault", "murdered" and "attack"), but CCD can make correct predictions based on the text features (Text) after causal intervention. In addition, after conducting counterfactual reasoning by subtracting the direct causal effect of the image (Image), the CCD is able to make correct predictions based on the debiased results. The two cases show the effectiveness of our CCD framework, which makes debiased predictions by removing the psycholinguistic bias in the text and image-only bias.

5 Related Work

In this section, we review the related work including fake news detection and causal inference.



(a) Fake news.



(b) Real news.

Figure 7: Two news cases from the Pheme dataset.

5.1 Multi-modal Fake News Detection

Existing fake news detection work generally falls into two categories: content-based methods and propagation-based methods. The multi-modal approaches fall into the former category.

Most works on multi-modal fake news detection exert efforts to fully incorporate cross-modal features. For instance, Jin et al. (2017) proposed a recurrent neural network with an attention mechanism to fuse the text, social context, and image features. Singhal et al. (2020) utilized pre-trained encoders and applied multiple-layer feature transformation to achieve deep fusion. Chen et al. (2022b) calculated the ambiguity score of different modalities to control the contribution of mono-modal features and inter-modal correlations to the final prediction. To capture fine-grained cross-modal correlations, Wu et al. (2021) employed multiple rounds of co-attention mechanism to model the cross-modal interactions. Qian et al. (2021b) lever-

aged a contextual attention network to model both the intra- and inter-modality information, and captured the hierarchical semantic information of the text. There are also methods leveraging external knowledge to provide powerful evidence or enrich features' representations (Hu et al., 2021; Qi et al., 2021). For example, Hu et al. (2021) compared each news with the external knowledge base through entities to utilize consistencies for detection.

In this work, we improve fake news detection from the perspective of causality and propose a novel framework that eliminates the hidden biases in each modality.

5.2 Causal Inference

Causal inference (Glymour et al., 2016) including causal intervention and counterfactual reasoning has been widely used in various fields such as recommendation (Zhang et al., 2021b; Wang et al., 2021), natural language inference (Tian et al., 2022), text classification (Qian et al., 2021a), named entity recognition (Zhang et al., 2021a), pretrained language models (Li et al., 2022), etc. It provides a powerful tool that can scientifically identify the causal correlations between variables and remove the hidden bias in the data. As for fake news detection, Zhu et al. (2022) eliminated the entity bias (the distribution of entities in the text) by counterfactual reasoning.

In this work, we discover the psycholinguistic bias and image-only bias in fake news detection, and propose a novel debiasing framework that eliminates these biases using causal intervention and counterfactual reasoning to enhance detection performance.

6 Conclusion

In this work, we propose a novel causal intervention and counterfactual reasoning based debiasing framework CCD that eliminates the hidden biases in multi-modal fake news detection. We analyze and identify the psycholinguistic bias in the text as well as the image-only bias. Then, we formulate the process of fake news detection as a causal graph, addressing the biases from the causality perspective. Specifically, we address the psycholinguistic bias by causal intervention with backdoor adjustment, and mitigate the image-only bias using counterfactual reasoning that subtracts the direct image-only causal effect from the total causal effect. Experiments on two real-world benchmark datasets verify that CCD can effectively eliminate biases and improve multi-modal fake news detection.

Limitations

When applying causal intervention to remove psycholinguistic bias, we utilize the LIWC dictionary to construct the confounder dictionary $\mathbf{D}_u$. We argue that the debiasing performance could be affected by the quality of the constructed confounder dictionary. In the future, we could try to improve the confounder dictionary with external knowledge.

Acknowledgements

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ACL 2023 Responsible NLP Checklist

A For every submission:

☑ A1. Did you describe the limitations of your work?
Section Limitation

☑ A2. Did you discuss any potential risks of your work?
Section Limitation

☑ A3. Do the abstract and introduction summarize the paper's main claims?
Abstract and Introduction

☒ A4. Have you used AI writing assistants when working on this paper?
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B ☑ Did you use or create scientific artifacts?

Section 3.2; Section 4.1

☑ B1. Did you cite the creators of artifacts you used?
Section 3.2; Section 4.1

☑ B2. Did you discuss the license or terms for use and / or distribution of any artifacts?
Section 3.2; Section 4.1

☑ B3. Did you discuss if your use of existing artifact(s) was consistent with their intended use, provided that it was specified? For the artifacts you create, do you specify intended use and whether that is compatible with the original access conditions (in particular, derivatives of data accessed for research purposes should not be used outside of research contexts)?
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The scientific articles used are provided with relevant documentation discussing this part.

☑ B5. Did you provide documentation of the artifacts, e.g., coverage of domains, languages, and linguistic phenomena, demographic groups represented, etc.?
Section1; Section 3.2; Section 4.1

☑ B6. Did you report relevant statistics like the number of examples, details of train / test / dev splits, etc. for the data that you used / created? Even for commonly-used benchmark datasets, include the number of examples in train / validation / test splits, as these provide necessary context for a reader to understand experimental results. For example, small differences in accuracy on large test sets may be significant, while on small test sets they may not be.
Section 4.1

C ☑ Did you run computational experiments?

Section 4

☐ C1. Did you report the number of parameters in the models used, the total computational budget (e.g., GPU hours), and computing infrastructure used?
Not applicable. Left blank.

The Responsible NLP Checklist used at ACL 2023 is adopted from NAACL 2022, with the addition of a question on AI writing assistance.

☑ C2. Did you discuss the experimental setup, including hyperparameter search and best-found hyperparameter values?
Section 4.1.4

☑ C3. Did you report descriptive statistics about your results (e.g., error bars around results, summary statistics from sets of experiments), and is it transparent whether you are reporting the max, mean, etc. or just a single run?
Section 4

☑ C4. If you used existing packages (e.g., for preprocessing, for normalization, or for evaluation), did you report the implementation, model, and parameter settings used (e.g., NLTK, Spacy, ROUGE, etc.)?
Section 4.1

D ☒ Did you use human annotators (e.g., crowdworkers) or research with human participants?

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☐ D1. Did you report the full text of instructions given to participants, including e.g., screenshots, disclaimers of any risks to participants or annotators, etc.?
No response.

☐ D2. Did you report information about how you recruited (e.g., crowdsourcing platform, students) and paid participants, and discuss if such payment is adequate given the participants' demographic (e.g., country of residence)?
No response.

☐ D3. Did you discuss whether and how consent was obtained from people whose data you're using/curating? For example, if you collected data via crowdsourcing, did your instructions to crowdworkers explain how the data would be used?
No response.

☐ D4. Was the data collection protocol approved (or determined exempt) by an ethics review board?
No response.

☐ D5. Did you report the basic demographic and geographic characteristics of the annotator population that is the source of the data?
No response.